Sprint 1 Database Table Analysis Report

Tables Analyzed: TVConfig and televisionReported

1. TVConfig Table

Purpose

The TVConfig table stores system configuration for TV content. It holds information about the host machine, timestamps, versions, directories, program IDs, and station slots.

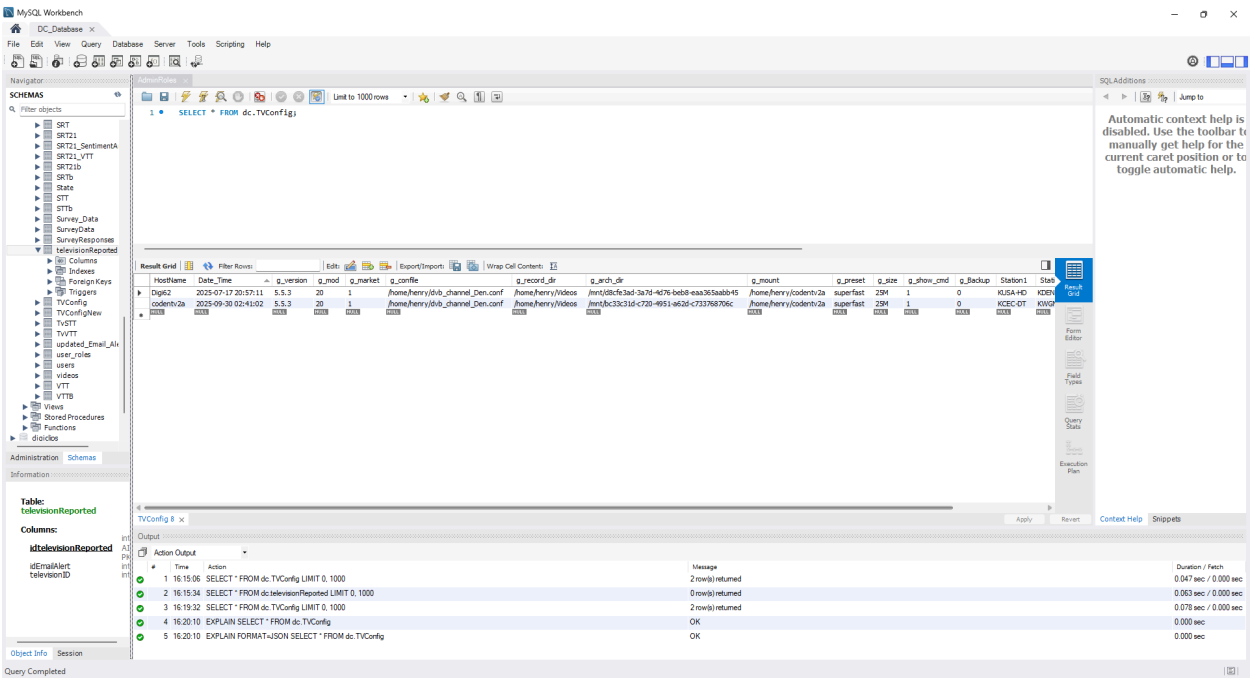

Table Structure

- **HostName** – Machine identifier
- **Date_Time** – Timestamp of record
- **g_version**, **g_mod**, **g_market**, **g_config** – Versioning and configuration parameters
- **g_record_dir**, **g_arch_dir** – Directory paths for recording and archiving
- **g_month**, **g_year**, **g_day** – Date parts
- **g_show_cmd** – Command string for shows
- **Station1 – Station7** – Station name slots

Indexes

- Internal **PRIMARY KEY** (system-managed).
- No explicit secondary indexes observed.

Observations / Issues

1. Only the primary index is defined; no optimization for frequent queries.
2. Important fields (e.g., Date_Time, HostName) do not enforce NOT NULL.

Recommendations

- Add indexes on Date_Time and g_market.
- Apply NOT NULL to required configuration fields.

2. televisionReported Table

Purpose

The televisionReported table logs flagged or reported TV clips, linking them back to a configuration.

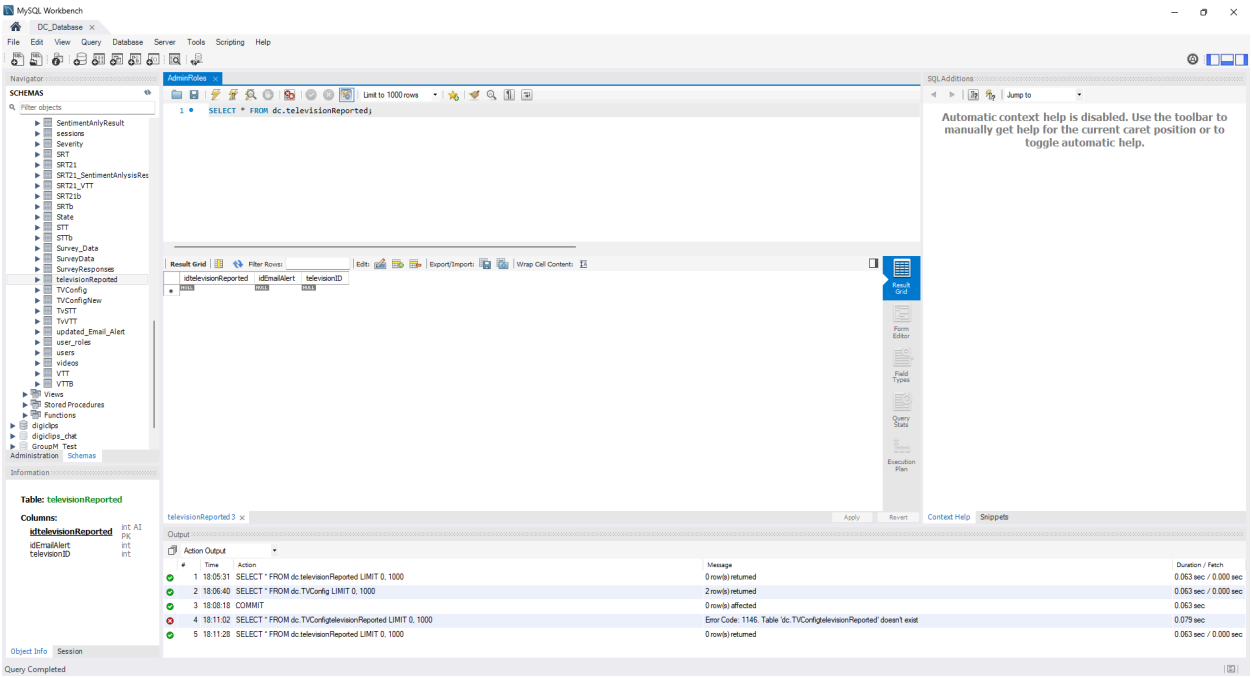

Table Structure

- **idtelevisionReported –** Unique identifier (Primary Key, Auto Increment).
- **idEmailAlert –** Reference to email alerts.
- **televisionID –** Reference to related TV entry.

Indexes

- **PRIMARY KEY on idtelevisionReported.**
- No defined foreign keys.

Observations / Issues

1. No enforced link between televisionReported.televisionID and TVConfig.
2. idEmailAlert is not enforced as a foreign key.
3. No NOT NULL constraints on key columns.

Recommendations

- Add foreign key: televisionReported.televisionID → TVConfig.
- Add foreign key: idEmailAlert → alerts table (if applicable).
- Apply NOT NULL to all key fields.

3. Relationship Mapping

- Intended relationship: televisionReported links to TVConfig.
- Type: One-to-Many (one configuration → many reports).
- Current state: Relationship not enforced in schema.

4. Testing

Testing was performed in MySQL Workbench using sample queries.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16:15:06	SELECT * FROM dc.TVConfig LIMIT 0, 1000	2 row(s) returned	0.047 sec / 0.000 sec
2	16:15:34	SELECT * FROM dc.televisionReported LIMIT 0, 1000	0 row(s) returned	0.063 sec / 0.000 sec

Query 1:

- Result: 2 row(s) returned.
- Status: Successful.

Query 2:

- Result: 0 row(s) returned.
- Status: Successful (empty table).

5. Summary

- TVConfig is functional but lacks constraints and indexes.
- televisionReported is empty at present and missing foreign key links.
- Both tables work but need schema improvements for long-term consistency and performance.